

6

Second Law of Thermodynamics

Learning Objectives

After reading this chapter you should be able to

- LO 6.1 Define the Second Law of thermodynamics and the difference between work and heat qualitatively
- LO 6.2 Identify differences between thermal and mechanical energy reservoirs
- LO 6.3 Explain heat engines in both closed and open systems
- LO 6.4 State Kelvin-Planck statement of the Second Law of thermodynamics
- LO 6.5 Describe working principle of refrigerator and heat pump from Clausius statement
- LO 6.6 Establish the equivalence of two statements of the Second Law of thermodynamics
- LO 6.7 Discuss reversible and irreversible processes and factors responsible for irreversibility
- LO 6.8 Discuss types of irreversibility
- LO 6.9 Examine condition for reversibility and the application of the most efficient cycle, i.e. Carnot cycle
- LO 6.10 Explain working of Carnot heat pump and refrigerator, and how Carnot theorem is applicable to validate its statement
- LO 6.11 Develop a point of view from Carnot's theorem and reach at the Third-Law of thermodynamics utilizing absolute thermodynamic temperature scale
- LO 6.12 Calculate the efficiency of reversible devices and provide the relation between ideal gas temperature and Kelvin temperature

6.1 INTRODUCTION TO THE SECOND LAW

The first law of thermodynamics states that a certain energy balance will hold when a system undergoes a change of state or a thermodynamic process. But it does not give any information on whether that change of state or the process is at all feasible or not. The first law cannot indicate whether a metallic bar of uniform temperature can spontaneously become warmer at one end and cooler at the other. All that the law can state is that if this process did occur, the energy gained by one end would be exactly equal to that lost by the other. It is the second law of thermodynamics which provides the criterion as to the probability of various processes.

Spontaneous processes in nature occur only in one direction. Heat always flows from a body at a higher temperature to a body at a lower temperature, fluid always flows from higher mechanical energy to mechanical energy. Material always diffuses from high concentration to low concentration. Time always flows in the forward direction. The reverse of these never happens spontaneously. The spontaneity of the process is due to a finite driving potential. These transfer processes can never occur spontaneously from a lower to a higher potential.

Figure 6.1 demonstrates the various transport processes. The process 1-2 (firm line) from a high to a low potential is spontaneous, whereas the process 2-1 (dotted line) can never happen spontaneously and is made possible only by energy addition.

The second law of thermodynamics puts a limitation on energy transformation other than that imposed by the first law.

Joule's experiments (Section 4.1) amply demonstrate that energy, when supplied to a system in the form of work, can be completely converted into heat (work transfer $\rightarrow$ internal energy increase $\rightarrow$ heat transfer). But the complete conversion of heat into work in a cycle is not possible. So heat and work are not completely interchangeable forms of energy.

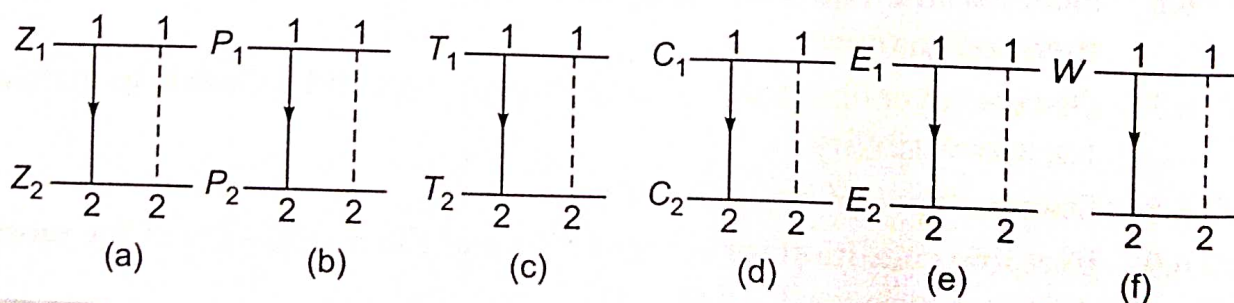


Fig. 6.1 (a) Flow from a high to a low elevation (Z), (b) Fluid flow from a high to a low pressure (p), (c) Heat transfer from a high to a low temperature (T), (d) Mass transfer of a constituent in a mixture from a high to a low concentration (C , gmol/m^3), (e) Flow of electric current from a high to a low electric potential (E), (f) Energy flux from work to heat transfer ($W \rightarrow Q$)

When work is converted into heat, we always have

$$W \equiv Q$$

but when heat is converted into work in a complete closed cycle process

$$Q \geq W$$

LO 6.1

Define the Second Law of thermodynamics and the difference between work and heat qualitatively

The arrow indicates the direction of energy transformation. This is illustrated in Fig. 6.2. As shown in Fig. 6.2(a), a system is taken from state 1 to state 2 by work transfer W_{1-2} , and then by heat transfer Q_{2-1} the system is brought back from state 2 to state 1 to complete a cycle. It is always found that $W_{1-2} = Q_{2-1}$. But if the system is taken from state 1 to state 2 by heat transfer Q_{1-2} , as shown in Fig. 6.2(b), then the system cannot be brought back from state 2 to state 1 by work transfer W_{2-1} . Hence, heat cannot be converted completely and continuously into work in a cycle. Some heat has to be rejected. In Fig. 6.2(c), Q_{1-2} is the heat supply to the system, W_{2-3} is the work done by the system, Q_{3-4} is the heat rejected from the system and W_{4-1} is the work done upon the system to complete the cycle. The net work done in the cycle is $(W_{2-3} - W_{4-1})$ which is less than Q_{1-2} and some heat Q_{3-4} must be rejected in the cycle. This underlies the work of Sadi Carnot, a French military engineer, who first studied this aspect of energy transformation (1824). Work is said to be a high grade energy and heat a low grade energy. The complete conversion of low grade energy into high grade energy in a cycle is impossible.

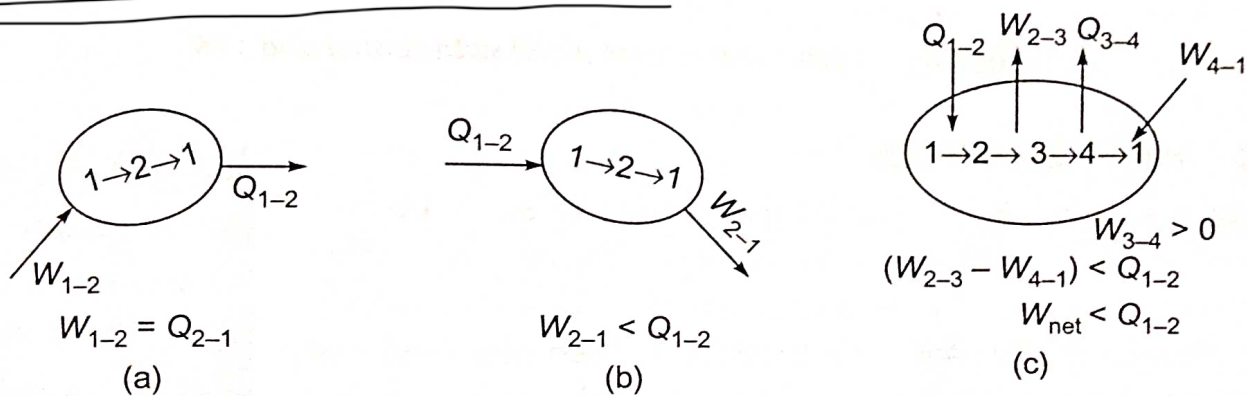


Fig. 6.2 Qualitative distinction between heat and work

6.2 ENERGY RESERVOIRS

A thermal energy reservoir (TER) is defined as a large body of infinite heat capacity, which is capable of absorbing or rejecting an unlimited quantity of heat without suffering appreciable changes in its thermodynamic coordinates. The changes that do take place in the large body as heat enters or leaves are so very slow and so very minute that all processes within it are quasi-static.

The thermal energy reservoir TER_H from which heat Q_1 is transferred to the system operating in a heat engine cycle is called the *source*. The thermal energy reservoir TER_L to which heat Q_2 is rejected from the system during a cycle is the *sink*. A typical source is a constant temperature furnace where fuel is continuously burnt, and a typical sink is a river or sea or the atmosphere itself.

A mechanical energy reservoir (MER) is a large body enclosed by an adiabatic impermeable wall capable of storing work as potential energy (such as a raised weight or wound spring) or kinetic energy (such as a rotating flywheel). All processes of interest within an MER are essentially quasi-static. An MER receives and delivers mechanical energy quasi-statically.

LO 6.2

Identify differences between thermal and mechanical energy reservoirs

The arrow indicates the direction of energy transformation. This is illustrated in Fig. 6.2. As shown in Fig. 6.2(a), a system is taken from state 1 to state 2 by work transfer W_{1-2} , and then by heat transfer Q_{2-1} the system is brought back from state 2 to state 1 to complete a cycle. It is always found that $W_{1-2} = Q_{2-1}$. But if the system is taken from state 1 to state 2 by heat transfer Q_{1-2} , as shown in Fig. 6.2(b), then the system cannot be brought back from state 2 to state 1 by work transfer W_{2-1} . Hence, heat cannot be converted completely and continuously into work in a cycle. Some heat has to be rejected. In Fig. 6.2(c), Q_{1-2} is the heat supply to the system, W_{2-3} is the work done by the system, Q_{3-4} is the heat rejected from the system and W_{4-1} is the work done upon the system to complete the cycle. The net work done in the cycle is $(W_{2-3} - W_{4-1})$ which is less than Q_{1-2} and some heat Q_{3-4} must be rejected in the cycle. This underlies the work of Sadi Carnot, a French military engineer, who first studied this aspect of energy transformation (1824). Work is said to be a *high grade energy* and heat a *low grade energy*. The complete conversion of low grade energy into high grade energy in a cycle is impossible.

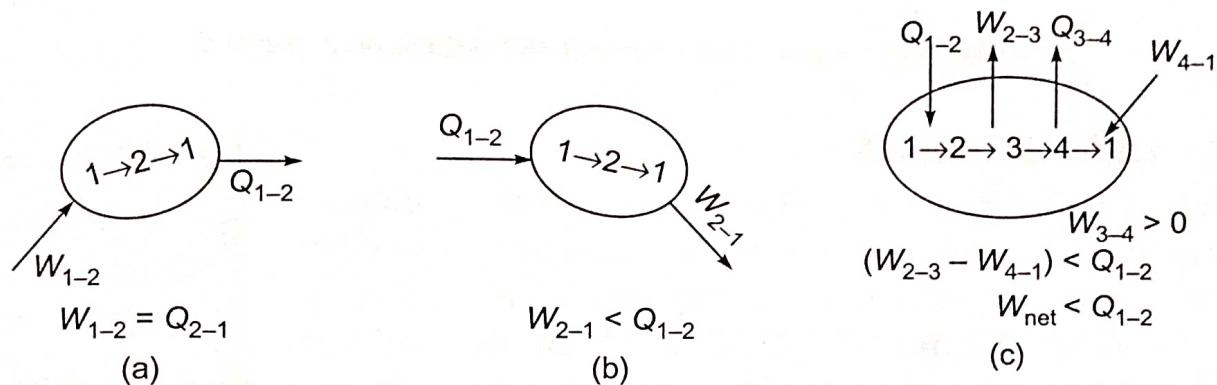


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LO 6.2

Identify differences between thermal and mechanical energy reservoirs

Figure 6.3 shows a cyclic heat engine exchanging heat with a source and a sink and delivering W_{net} in a cycle to an MER.

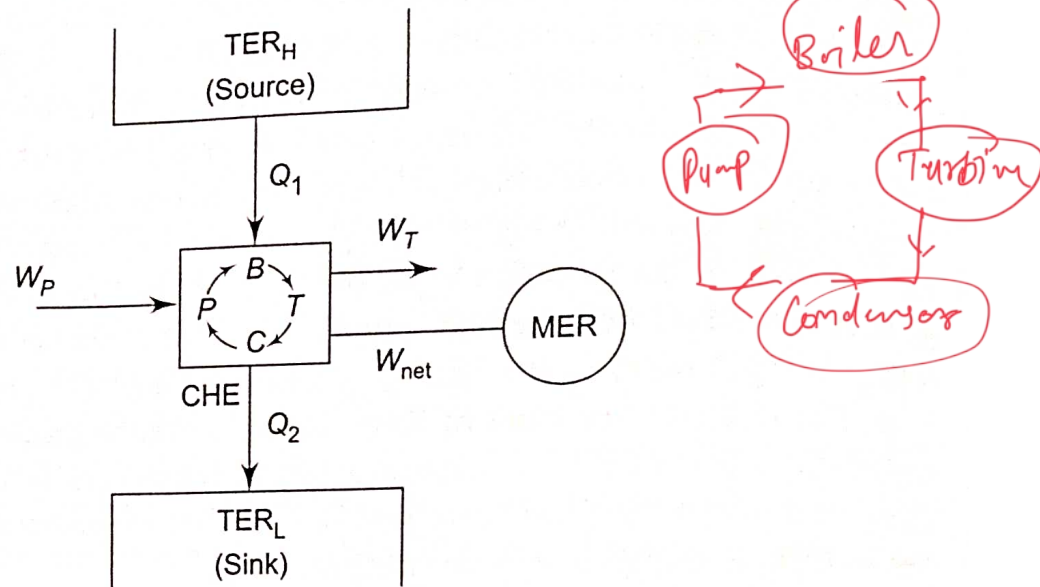


Fig. 6.3 Cyclic heat engine (CHE) with source and sink

6.3 HEAT ENGINES

It has been already mentioned that work can easily be converted to other forms of energy; however, conversion of other forms of energy to work is not an easy task. From the foregoing discussion (Section 6.1), it has also been observed that work can be converted to heat directly and completely, but complete conversion of heat to work is not possible. For the purpose of conversion of heat into work, we require special devices, called heat engines.

A heat engine may be defined as a device, which operates in a thermodynamic cycle, converts part of the heat received from high temperature reservoir to useful work and rejects the remaining heat to low temperature reservoir. Thus, there is a net heat transfer to a heat engine and a network transfer from the heat engine. The thermodynamic cycle in which the heat engine works is called heat engine cycle. Heat engines usually involve a substance which receives energy from the source and rejects energy to the sink. This substance is called the working fluid or working substance.

A heat engine may be in the form of mass of gas in a cylinder bounded by a piston (Fig. 6.4a) or a mass of water moving in a steady flow through a steam power plant (Fig. 6.4b). In the cyclic heat engine, as represented in Fig. 6.4(a), heat Q_1 is transferred to the system, work W_E is done by the system, work W_C is done upon the system, and then heat Q_2 is rejected from the system. The system is brought back to the initial state through these four successive processes which constitute a heat engine cycle. In Fig. 6.4(b), heat Q_1 is transferred from the furnace to the water in the boiler to form steam which then works on the turbine rotor to produce work W_T , then the steam is condensed to water in the condenser in which an amount of heat Q_2 is rejected from the system, and finally work W_P is done on the system (water) to pump it to the boiler. The system repeats the cycle.

LO 6.3

Explain heat engines in both closed and open systems

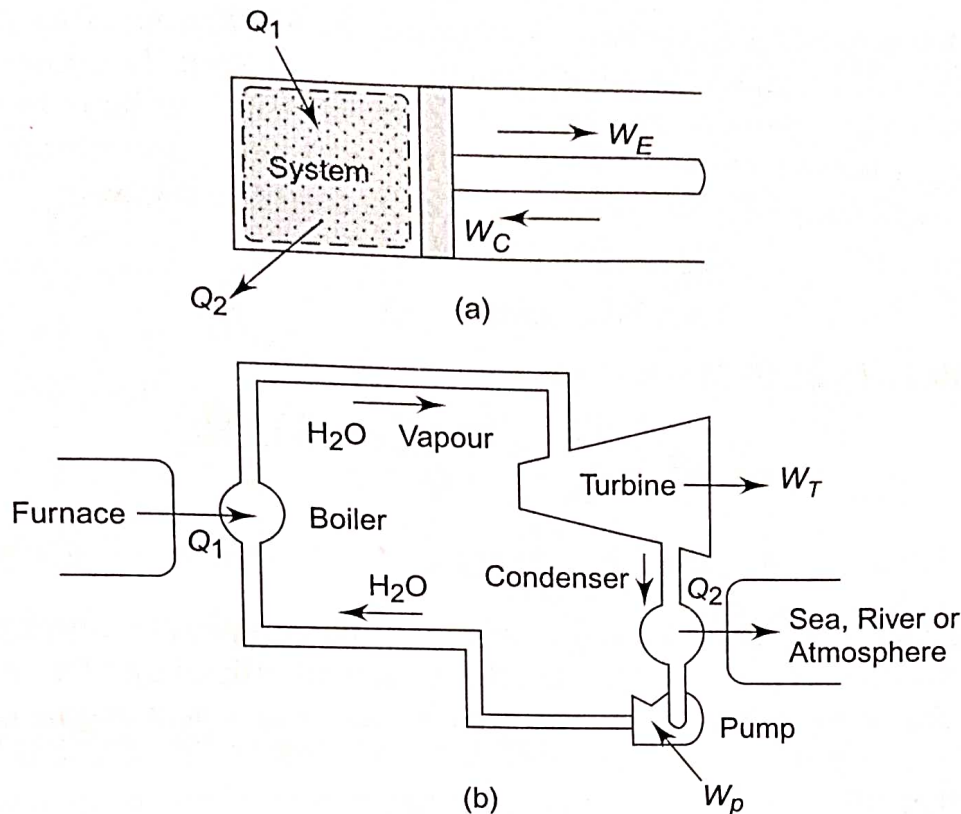


Fig. 6.4 *Cyclic heat engine (a) Heat engine cycle performed by a closed system undergoing four successive energy interactions with the surroundings, (b) Heat engine cycle performed by a steady flow system interacting with the surroundings as shown*

The net heat transfer in a cycle to either of the heat engines is given as:

$$Q_{\text{net}} = Q_1 - Q_2 \quad (6.1)$$

and the net work transferred in a cycle is given as:

$$W_{\text{net}} = W_T - W_P \quad (6.2)$$

(or $W_{\text{net}} = W_E - W_C$)

By the first law of thermodynamics for a cycle, we have:

$$\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W$$

$\therefore Q_{\text{net}} = W_{\text{net}}$

or $Q_1 - Q_2 = W_T - W_P \quad (6.3)$

Figure 6.5 represents a cyclic heat engine in the form of a block diagram indicating various energy interactions during a cycle. Boiler (B), turbine (T), condenser (C), and pump (P), all four together constitute a heat engine. A heat engine here is a certain quantity of water undergoing the energy interactions, as shown, in cyclic operations to produce net work from a certain heat input.

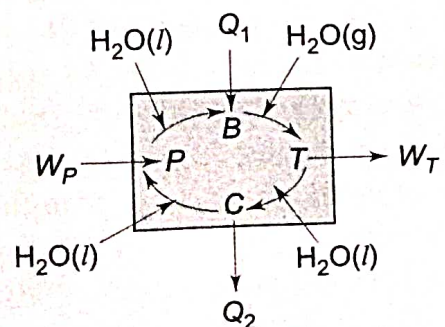


Fig. 6.5 *Cyclic heat engine with energy interactions represented in a block diagram*

The performance of a heat engine is expressed by the thermal efficiency, which is defined as the ratio of the desired output to the required input. The function of a heat engine is to produce work *continuously* at the expense of heat input to it, hence the net work W_{net} (desired output) and heat Q_1 (required input) are of primary interest. The thermal efficiency of a heat engine can be expressed as follows:

$$\eta = \frac{\text{Net work output}}{\text{Total heat input}} = \frac{W_{\text{net}}}{Q_1} \quad (6.4)$$

From Eqs. (6.1), (6.2), (6.3) and (6.4),

$$\eta = \frac{W_{\text{net}}}{Q_1} = \frac{W_T - W_P}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

or

$$\eta = 1 - \frac{Q_2}{Q_1} \quad (6.5)$$

✗ Since heat Q_1 transferred to a system cannot be completely converted to work in a cycle (Section 6.1), $W_{\text{net}} < Q_1$. Therefore, thermal efficiency of a heat engine is always less than unity. Alternatively, we can conclude that a heat engine can never be 100% efficient.

The thermal efficiency of a heat engine is a parameter that measures how efficiently it converts the heat received to work. The thermal efficiencies of real engines are relatively low. It is important to know in this context the typical values of the thermal efficiency of real engines. The usual values of the thermal efficiency are about 30–35% for spark ignition automobile engines, 35–40% for diesel engines, 35–50% for large power plants, and 60–65% for modern gas-steam power plants. Therefore, a significant amount of energy supplied to the engine is useless and this motivates the engineers to work on engines to improve their thermal efficiencies.

6.4 KELVIN-PLANCK STATEMENT OF THE SECOND LAW

It is clear from the previous discussion that no heat engine can convert all the heat it receives to useful work. In other words, a heat engine, operating in a cycle, must reject some heat to a low-temperature reservoir. To produce net work in a thermodynamic cycle, a heat engine has thus to exchange heat with two reservoirs, the source and the sink. This limitation forms the basis of the *Kelvin-Planck statement* of the second law of thermodynamics, which is expressed as follows: It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.

If $Q_2 = 0$ (i.e., $W_{\text{net}} = Q_1$, or $\eta = 1.00$), the heat engine will produce net work in a complete cycle by exchanging heat with only one reservoir, thus violating the Kelvin-Planck statement (Fig. 6.6). Such a heat engine is called a perpetual motion machine of the second kind, abbreviated to PMM2. A PMM2 is impossible.

A heat engine has, therefore, to exchange heat with two thermal energy reservoirs at two different temperatures to produce net work in a complete cycle (Fig. 6.7). So long as there is a difference in temperature, motive power (i.e., work) can be produced. If the bodies with which the heat engine exchanges heat are of finite heat capacities, work will be produced by the heat engine till the temperatures of the two bodies are equalised.

LO 6.4

State Kelvin-Planck statement of the Second Law of thermodynamics

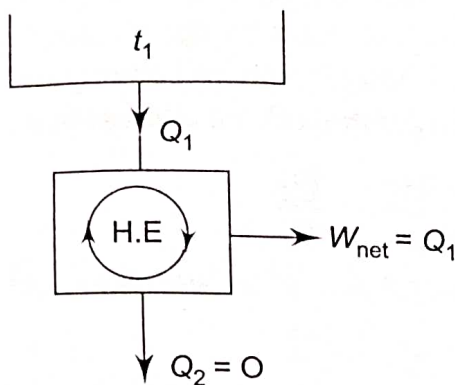


Fig. 6.6 A PMM2

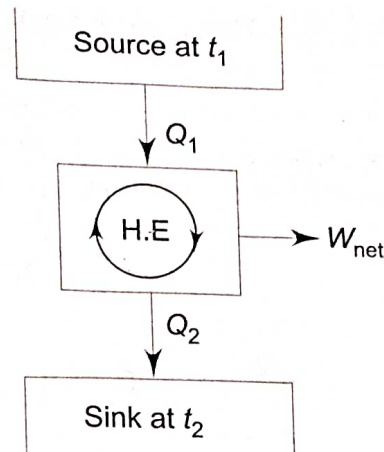


Fig. 6.7

Heat engine producing net work in a cycle by exchanging heat at two different temperatures

6.5 CLAUSIUS' STATEMENT OF THE SECOND LAW

Heat always flows from a body at a higher temperature to a body at a lower temperature. The reverse process never occurs spontaneously.

The above observation forms the basis of the Clausius statement of the second law of thermodynamics, which is as follows: It is impossible to construct a device which, operating in a cycle, will produce no effect other than the transfer of heat from a cooler to a hotter body.

Heat cannot flow of itself from a body at a lower temperature to a body at a higher temperature. Some work must be expended to achieve this.

6.6 REFRIGERATOR AND HEAT PUMP

A refrigerator is a device which, operating in a cycle, maintains a body at a temperature lower than the temperature of the surroundings. Let the body *A* (Fig. 6.8) be maintained at t_2 , which is lower than the ambient temperature t_1 . Even though *A* is insulated, there will always be heat leakage Q_2 into the body from the surroundings by virtue of the temperature difference. In order to maintain body *A* at the constant temperature t_2 , heat has to be removed from the body at the same rate at which heat is leaking into the body. This heat (Q_2) is absorbed by a working fluid, called the *refrigerant*, which evaporates in the evaporator E_1 at a temperature lower than t_2 absorbing the latent heat of vaporisation from the body *A* which is cooled or refrigerated (Process 4–1). The vapour is first compressed in the compressor C_1 , driven by a motor, which absorbs work W_C (Process 1–2), and is then condensed in the condenser C_2 rejecting the latent heat of condensation Q_1 at a temperature higher than that of the atmosphere (at t_1) for heat transfer to take place (Process 2–3). The condensate then expands adiabatically through an expander (an engine or turbine) producing work W_E , when the temperature drops to a value lower than t_2 such that heat Q_2 flows from the body *A* to make the refrigerant evaporate (Process 3–4). Such a cyclic device of flow through E_1 – C_1 – C_2 – E_2 is called a *refrigerator*. Just

LO 6.5

Describe working principle of refrigerator and heat pump from Clausius statement

like thermal efficiency in a heat engine, the parameter that characterises the performance of a refrigerator is called the *coefficient of performance*, abbreviated to COP, which is defined as the ratio of the desired effect to the required input.

In a refrigerator cycle removal of heat Q_2 from a body A (desired effect) and work input W are of primary interest. The COP of a refrigerator can be expressed as

$$\text{COP} = \frac{\text{Desired effect}}{\text{Work input}} = \frac{Q_2}{W}$$

$$\therefore [\text{COP}]_{\text{ref}} = \frac{Q_2}{Q_1 - Q_2} \quad (6.6)$$

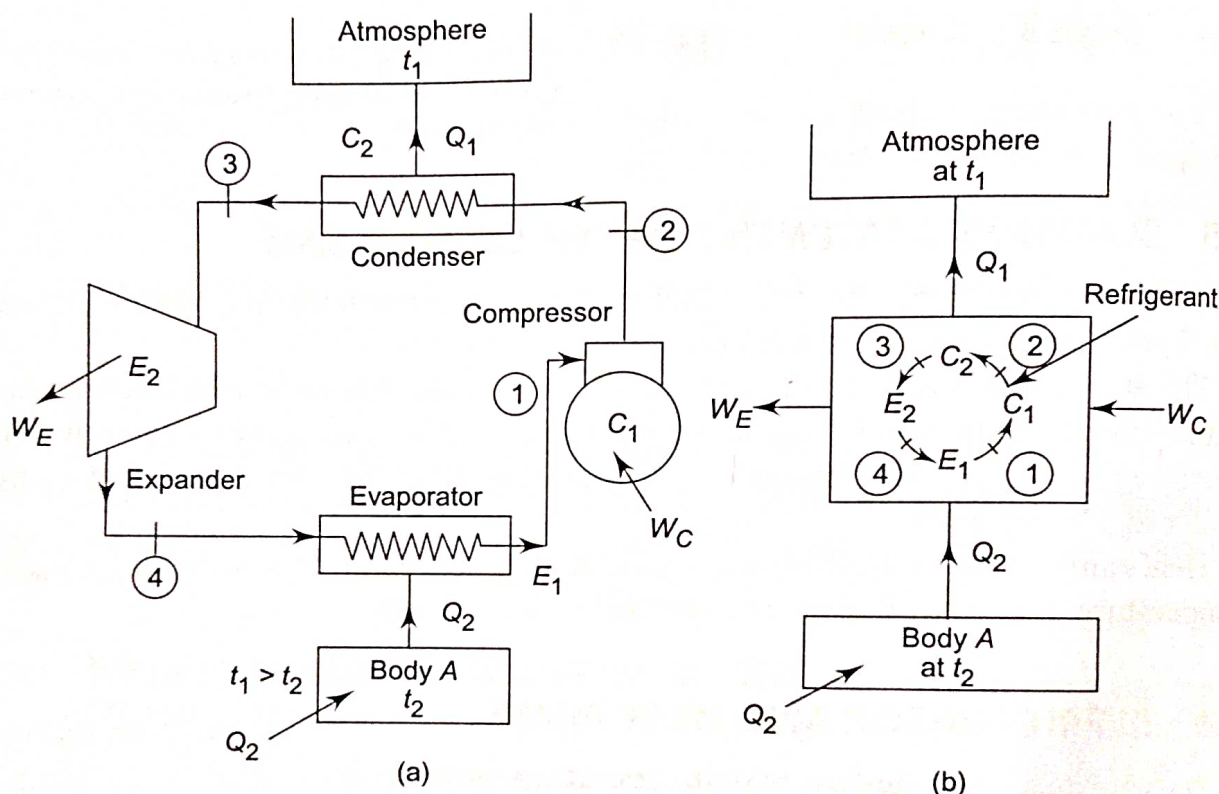


Fig. 6.8 A cyclic refrigeration plant

A heat pump is a device which, operating in a cycle, maintains a body, say B (Fig. 6.9), at a temperature higher than the temperature of the surroundings. By virtue of the temperature difference, there will be heat leakage Q_1 from the body to the surroundings. The body will be maintained at the constant temperature t_1 , if heat is discharged into the body at the same rate at which heat leaks out of the body. The heat is extracted from the low temperature reservoir, which is nothing but the atmosphere, and discharged into the high temperature body B , with the expenditure of work W in a cyclic device called a heat pump. The working fluid operates in a

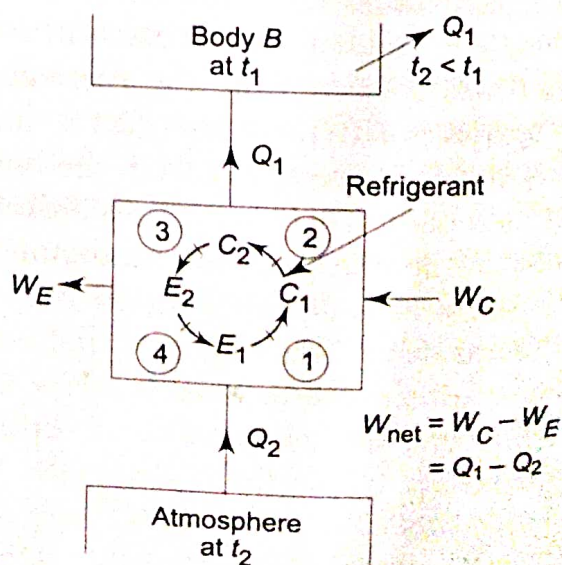


Fig. 6.9 A cyclic heat pump

cycle flowing through the evaporator E_1 , compressor C_1 , condenser C_2 and expander E_2 , similar to a refrigerator, but the attention is here focussed on the high temperature body B . Here Q_1 and W are of primary interest, and the COP is defined as

$$\text{COP} = \frac{Q_1}{W}$$

$$\therefore [\text{COP}]_{H.P.} = \frac{Q_1}{Q_1 - Q_2} \quad (6.7)$$

From Eqs. (6.6) and (6.7), it is found that

$$[\text{COP}]_{H.P.} = [\text{COP}]_{\text{ref}} + 1 \quad (6.8)$$

The COP of a heat pump is greater than the COP of a refrigerator by unity. For heat to flow from a cooler to a hotter body, W cannot be zero, and hence, the COP (both for refrigerator and heat pump) cannot be infinity. Therefore, $W > 0$, and $\text{COP} < \infty$.

It may be noted that the energy conversion efficiency from energy input (work) to desired energy output (heat) for a refrigerator or heat pump as expressed by Eqs. (6.6) and (6.7), exceeds 100% without violating any law. However, this is in sharp contrast to the thermal efficiency which can never be greater than 100%. To eliminate this confusion, the expression 'coefficient of performance' (COP) is used instead of 'thermal efficiency' in such situations.

6.6.1 Heat Pump vs Electric Resistance Heater

For an electrical resistance heater, if W is the electrical energy consumption, then the heat transferred to the space at steady state is W only, i.e., $Q_1 = W$.

When electrical energy W is used to drive the compressor of a heat pump, the heat supplied to the space will be

$$Q_1 = [\text{COP}]_{\text{HP}} W$$

Using Eq. (6.8), we have

$$Q_1 = [\text{COP}_{\text{ref}} + 1] W \quad (6.9)$$

Since $[\text{COP}]_{\text{ref}}$ is a positive quantity, it is evident from Eq. (6.9) that $Q_1 > W$. This implies that the heat supplied by a heat pump Q_1 will always be more than the energy consumption W .

The above observation can be demonstrated through an example as follows: A 1 kW electric heater can give 1 kW of heat at steady state and nothing more. On the other hand, when 1 kW electrical energy is used to drive the compressor of a heat pump, the heat supplied will be always more than 1 kW. This shows that the use of electric heater for the purpose of heating is thermodynamically inefficient.

6.7 EQUIVALENCE OF KELVIN-PLANCK AND CLAUSIUS STATEMENTS

At first sight, Kelvin-Planck's and Clausius' statements may appear to be unconnected, but it can easily be shown that they are virtually two parallel statements of the second law and are equivalent in all respects.

LO 6.6

Establish the equivalence of two statements of the Second Law of thermodynamics

The equivalence of the two statements will be proved if it can be shown that the violation of one statement implies the violation of the second, and vice versa.

1. Let us first consider a cyclic heat pump P which transfers heat from a low temperature reservoir (t_2) to a high temperature reservoir (t_1) with no other effect, i.e., with no expenditure of work, violating Clausius statement (Fig. 6.10).

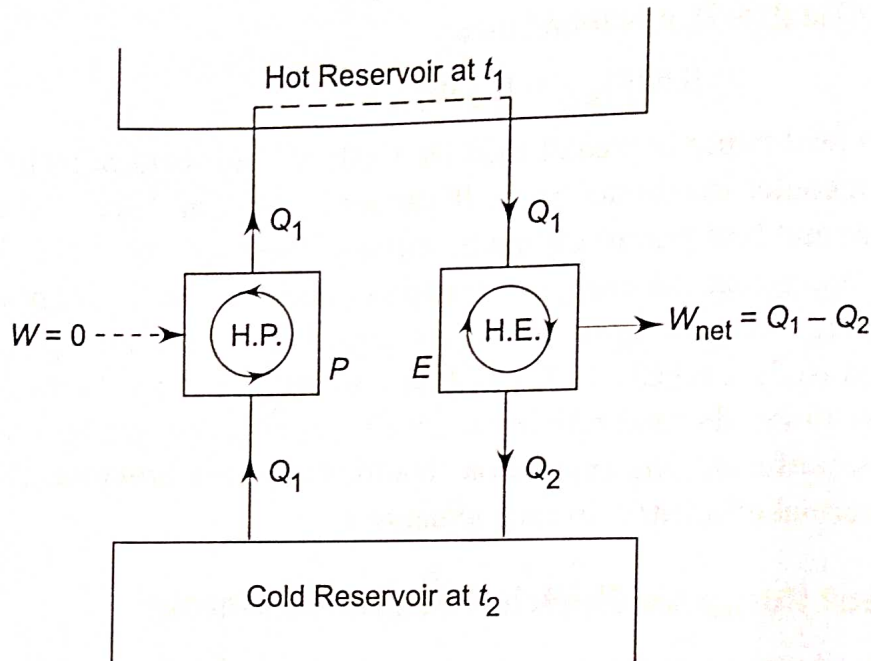


Fig. 6.10 Violation of the Clausius statement

Let us assume a cyclic heat engine E operating between the same thermal energy reservoirs, producing W_{net} in one cycle. The rate of working of the heat engine is such that it draws an amount of heat Q_1 from the hot reservoir equal to that discharged by the heat pump. Then the hot reservoir may be eliminated and the heat Q_1 discharged by the heat pump is fed to the heat engine. So we see that the heat pump P and the heat engine E acting together constitute a heat engine operating in cycles and producing net work while exchanging heat only with one body at a single fixed temperature (t_2). This violates the Kelvin-Planck statement.

2. Let us now consider a perpetual motion machine of the second kind (E) which produces net work in a cycle by exchanging heat with only one thermal energy reservoir (at t_1) and thus violates the Kelvin-Planck statement (Fig. 6.11).

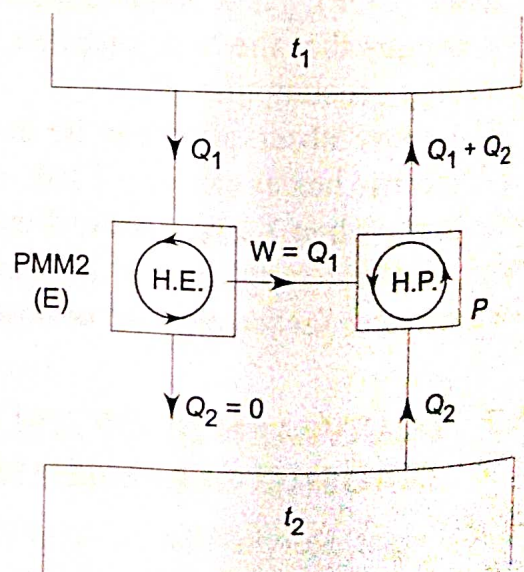


Fig. 6.11

Violation of the Kelvin-Planck statement

Let us assume a cyclic heat pump (P) extracting heat Q_2 from a low temperature reservoir at t_2 and discharging heat to the high temperature reservoir at t_1 with the expenditure of work W equal to what the PMM2 delivers in a complete cycle. So E and P together constitute a heat pump working in cycles and producing the sole effect of transferring heat from a lower to a higher temperature body, thus violating the Clausius statement.

6.8 REVERSIBILITY AND IRREVERSIBILITY

The second law of thermodynamics enables us to divide all processes into two classes:

1. Reversible or ideal process
2. Irreversible or natural process

A reversible process is one which is performed in such a way that at the conclusion of the process, both the system and the surroundings may be restored to their initial states, without producing any changes in the rest of the universe. Let the state of a system be represented by A (Fig. 6.12), and let the system be taken to state B quasistatically by following the path $A-B$. If the system and also the surroundings are restored to their initial states on reverse, and no change in the universe is produced, then the process $A-B$ will be a reversible process. In the reverse process, the system has to be taken from state B to A by following the same path $B-A$. A reversible process should not leave any trace or relic behind to show that the process had ever occurred.

A reversible process is carried out infinitely slowly with an infinitesimal gradient, so that every state passed through by the system is an equilibrium state. So a reversible process coincides with a quasi-static process.

Any natural process carried out with a finite gradient is an irreversible process. A reversible process, which consists of a succession of equilibrium states, is an idealised hypothetical process, approached only as a limit. It is said to be an asymptote to reality. All spontaneous processes are irreversible.

Time has an important effect on reversibility. If the time allowed for a process to occur is infinitely large, even though the gradient is finite, the process becomes reversible. However, if this time is squeezed to a finite value, the finite gradient makes the process irreversible.

6.9 CAUSES OF IRREVERSIBILITY

The irreversibility of a process may be due to either one or both of the following:

1. Lack of equilibrium during the process
2. Involvement of dissipative effects

LO 6.7

Discuss reversible and irreversible processes and factors responsible for irreversibility

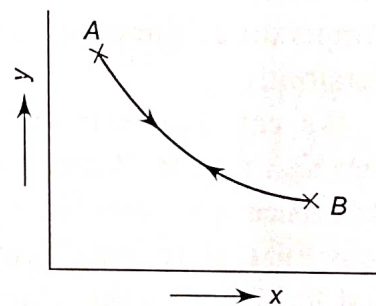


Fig. 6.12 A reversible process

6.9.1 Irreversibility due to Lack of Equilibrium

The lack of equilibrium (mechanical, thermal or chemical) between the system and its surroundings, or between two systems, or two parts of the same system, causes a spontaneous change which is irreversible. The following are specific examples in this regard:

1. Heat transfer through a finite temperature difference A heat transfer process approaches reversibility as the temperature difference between two bodies approaches zero. We define a reversible heat transfer process as one in which heat is transferred through an infinitesimal temperature difference. So to transfer a finite amount of heat through an infinitesimal temperature difference would require an infinite amount of time, or infinite area. All actual heat transfer processes are through a finite temperature difference and are, therefore, irreversible, and greater the temperature difference, the greater is the irreversibility.

We can demonstrate by the second law that heat transfer through a finite temperature difference is irreversible. Let us assume that a source at t_A and a sink at t_B ($t_A > t_B$) are available, and let Q_{A-B} be the amount of heat flowing from A to B (Fig. 6.13). Let us assume an engine operating between A and B , taking heat Q_1 from A and discharging heat Q_2 to B . Let the heat transfer process be reversed, and Q_{B-A} be the heat flowing from B to A , and let the rate of working of the engine be such that

$$Q_2 = Q_{B-A}$$

(Fig. 6.14). Then the sink B may be eliminated. The net result is that E produces network W in a cycle by exchanging heat only with A , thus violating the Kelvin-Planck statement. So the heat transfer process Q_{A-B} is irreversible, and Q_{B-A} is not possible.

2. Lack of pressure equilibrium within the interior of the system or between the system and the surroundings When there exists a difference in pressure between the

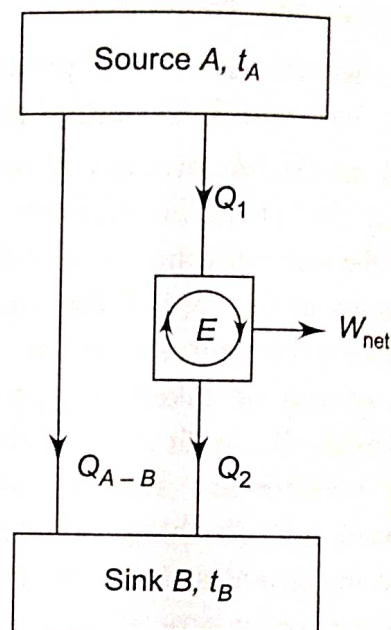


Fig. 6.13 Heat transfer through a finite temperature difference

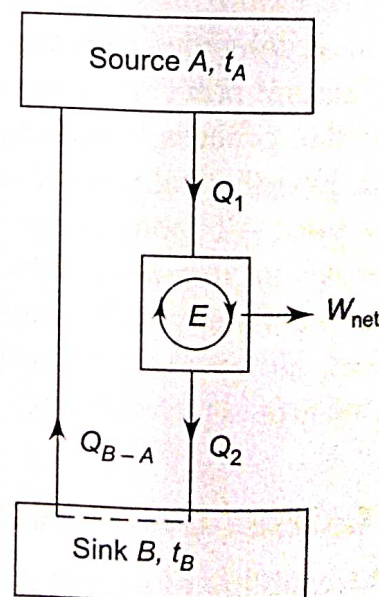


Fig. 6.14 Heat transfer through a finite temperature difference is irreversible

system and the surroundings, or within the system itself, then both the system and its surroundings or the system alone, will undergo a change of state which will cease only when mechanical equilibrium is established. The reverse of this process is not possible spontaneously without producing any other effect. That the reverse process will violate the second law becomes obvious from the following illustration.

3. Free expansion Let us consider an insulated container (Fig. 6.15) which is divided into two compartments A and B by a thin diaphragm. Compartment A contains a mass of gas, while compartment B is completely evacuated. If the diaphragm is punctured, the gas in A will expand into B until the pressures in A and B become equal. This is known as *free* or *unrestrained expansion*. We can demonstrate by the second law, that the process of free expansion is irreversible.

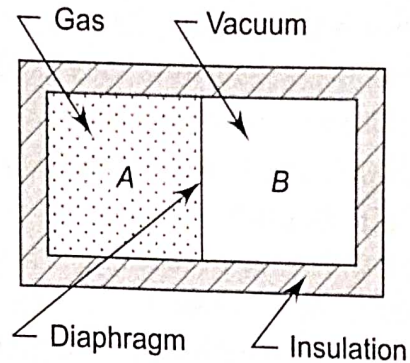


Fig. 6.15 Free expansion

To prove this, let us assume that free expansion is reversible, and that the gas in B returns into A with an increase in pressure, and B becomes evacuated as before (Fig. 6.16). There is no other effect. Let us install an engine (a machine, not a cyclic heat engine) between A and B , and permit the gas to expand through the engine from A to B . The engine develops a work output W at the expense of the internal energy of the gas. The internal energy of the gas (system) in B can be restored to its initial value by heat transfer Q ($= W$) from a source. Now, by the use of the reversed free expansion, the system can be restored to the initial state of high pressure in A and vacuum in B . The net result is a cycle, in which we observe that net work output W is accomplished by exchanging heat with a single reservoir. This violates the Kelvin–Planck statement. Hence, free expansion is irreversible.

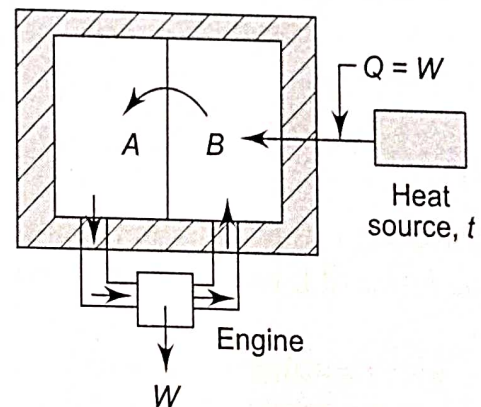


Fig. 6.16 Second law demonstrates that free expansion is irreversible

The same argument will hold if the compartment B is not in vacuum but at a pressure lower than that in compartment A (case b).

6.9.2 Irreversibility due to Dissipative Effects

The irreversibility of a process may be due to the *dissipative effects* in which work is done without producing an equivalent increase in the kinetic or potential energy of any system. The transformation of work into molecular internal energy either of the system or of the reservoir takes place through the agency of such phenomena as friction, viscosity, inelasticity, electrical resistance and magnetic hysteresis. These effects are known as dissipative effects, and work is said to be dissipated.

1. Friction Friction is always present in moving devices. Friction may be reduced by suitable lubrication, but it can never be completely eliminated. If this were possible, a movable device could be kept in a continual motion without violating either of the two laws of thermodynamics. The continual motion of a movable device in the complete absence of friction is known as *perpetual motion of the third kind*.

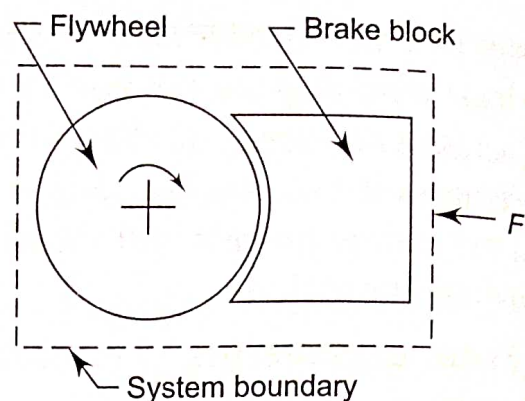


Fig. 6.17 Irreversibility due to dissipative effect like friction

That friction makes a process irreversible can be demonstrated by the second law. Let us consider a system consisting of a flywheel and a brake block (Fig. 6.17). The flywheel was rotating with a certain rpm, and it was brought to rest by applying the friction brake. The distance moved by the brake block is very small, so work transfer is very nearly equal to zero. If the braking process occurs very rapidly, there is a little heat transfer. Using suffix 2 after braking and suffix 1 before braking, and applying the first law, we have

$$\begin{aligned}
 Q_{1-2} &= E_2 - E_1 + W_{1-2} \\
 0 &= E_2 - E_1 + 0 \\
 \therefore E_2 &= E_1
 \end{aligned} \tag{6.10}$$

The energy of the system (isolated) remains constant. Since the energy may exist in the forms of kinetic, potential and molecular internal energy, we have

$$U_2 + \frac{mV_2^2}{2} + mZ_2g = U_1 + \frac{mV_1^2}{2} + mZ_1g$$

Since the wheel is brought to rest, $V_2 = 0$, and there is no change in P.E.

$$U_2 = U_1 + \frac{mV_1^2}{2} \tag{6.11}$$

Therefore, the molecular internal energy of the system (i.e. of the brake and the wheel) increases by the absorption of the K.E. of the wheel. The reverse process, i.e. the conversion of this increase in molecular internal energy into K.E. within the system to cause the wheel to rotate is not possible. To prove it by the second law, let us assume that it is possible, and imagine the following cycle with three processes:

Process A Initially, the wheel and the brake are at high temperature as a result of the absorption of the K.E. of the wheel, and the flywheel is at rest. Let the flywheel now start rotating at a particular rpm at the expense of the internal energy of the wheel and brake, the temperature of which will then decrease.

Process B Let the flywheel be brought to rest by using its K.E. in raising weights, with no change in temperature.

Process C Now, let heat be supplied from a source to the flywheel and the brake to restore the system to its initial state.

Therefore, the processes *A*, *B* and *C* together constitute a cycle producing work by exchanging heat with a single reservoir. This violates the Kelvin–Planck statement, and it will become a PMM2. So the braking process, i.e. the transformation of K.E. into molecular internal energy, is irreversible.

2. Paddle-Wheel work transfer Work may be transferred into a system in an insulated container by means of a paddle wheel (Fig. 6.18) which is also known as stirring work. Here work transferred is dissipated adiabatically into an increase in the molecular internal energy of the system. To prove the irreversibility of the process, let us assume that the same amount of work is delivered by the system at the expense of its molecular internal energy, and the temperature of the system goes down (Fig. 6.19). The system is brought back to its initial state by heat transfer from a source. These two processes together constitute a cycle in which there is work output and the system exchanges heat with a single reservoir. It becomes a PMM2, and hence the dissipation of stirring work to internal energy is irreversible.

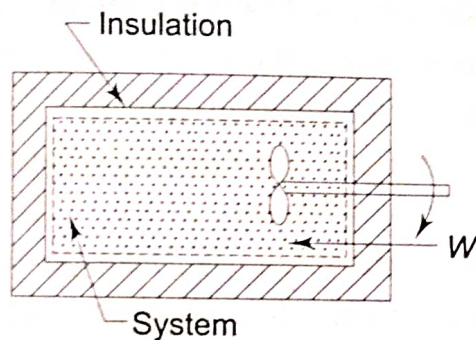


Fig. 6.18 Adiabatic work transfer

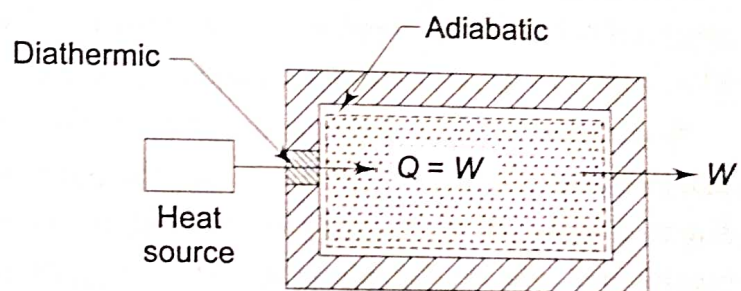


Fig. 6.19 Irreversibility due to dissipation of stirring work into internal energy

3. Transfer of electricity through a resistor

The flow of electric current through a wire represents work transfer, because the current can drive a motor which can raise a weight. Taking the wire or the resistor as the system (Fig. 6.20) and writing the first law

$$Q_{1-2} = U_2 - U_1 + W_{1-2}$$

Here, both W_{1-2} and Q_{1-2} are negative.

$$W_{1-2} = U_2 - U_1 + Q_{1-2}$$

(6.12)

A part of the work transfer is stored as an increase in the internal energy of the wire (to give an increase in its temperature), and the remainder leaves the system as heat. At steady state, the internal energy and hence the temperature of the resistor become constant with respect to time and

$$W_{1-2} = Q_{1-2}$$

(6.13)

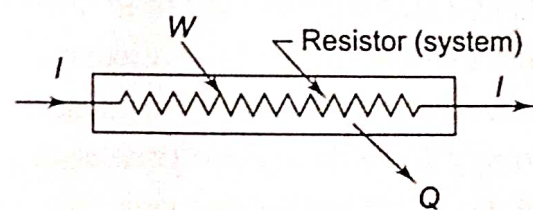


Fig. 6.20 Irreversibility due to dissipation of electrical work into internal energy

The reverse process, i.e. the conversion of heat Q_{1-2} into electrical work W_{1-2} of the same magnitude is not possible. Let us assume that this is possible. Then heat Q_{1-2} will be absorbed and equal work W_{1-2} will be delivered. But this will become a PMM2. So the dissipation of electrical work into internal energy or heat is irreversible.

6.10 TYPES OF IRREVERSIBILITY

It has been discussed in Section 6.9 that a process becomes irreversible if it occurs due to a finite potential gradient like the gradient in temperature or pressure, or if there is dissipative effect like friction, in which work is transformed into internal energy increase of the system. Two types of irreversibility can be distinguished:

1. Internal irreversibility
2. External irreversibility

The internal irreversibility is caused by the internal dissipative effects like friction, turbulence, electrical resistance, magnetic hysteresis, etc., within the system. The external irreversibility refers to the irreversibility occurring at the system boundary like heat interaction with the surroundings due to a finite temperature gradient.

Sometimes, it is useful to make other distinctions. If the irreversibility of a process is due to the dissipation of work into the increase in internal energy of a system, or due to a finite pressure gradient, it is called *mechanical irreversibility*. If the process occurs on account of a finite temperature gradient, it is *thermal irreversibility*, and if it is due to a finite concentration gradient or a chemical reaction, it is called *chemical irreversibility*.

A heat engine cycle in which there is a temperature difference (i) between the source and the working fluid during heat supply, and (ii) between the working fluid and the sink during heat rejection exhibits external thermal irreversibility. If the real source and sink are not considered and hypothetical reversible processes for heat supply and heat rejection are assumed, the cycle can be reversible. With the inclusion of the actual source and sink, however, the cycle becomes externally irreversible.

6.11 CONDITIONS FOR REVERSIBILITY

A natural process is irreversible because the conditions for mechanical, thermal and chemical equilibrium are not satisfied, and the dissipative effects, in which work is transformed into an increase in internal energy, are present. For a process to be reversible, it must not possess these features. If a process is performed

LO 6.8

Discuss types of irreversibility

LO 6.9

Examine condition for reversibility and the application of the most efficient cycle, i.e. Carnot cycle

quasistatically, the system passes through states of thermodynamic equilibrium, which may be traversed as well in one direction as in the opposite direction. *If there are no dissipative effects, all the work done by the system during the performance of a process in one direction can be returned to the system during the reverse process.*

A process will be reversible when it is performed in such a way that the system is at all times infinitesimally near a state of thermodynamic equilibrium and in the absence of dissipative effect of any form. Reversible processes are, therefore, purely ideal, limiting cases of actual processes.

6.12 CARNOT CYCLE

A reversible cycle is an ideal hypothetical cycle in which all the processes constituting the cycle are reversible. Carnot cycle is a reversible cycle. For a stationary system, as in a piston and cylinder machine, the cycle consists of the following four successive processes (Fig. 6.21):

1. *A reversible isothermal process* in which heat Q_1 enters the system at t_1 reversibly from a constant temperature source at t_1 when the cylinder cover is in contact with the diathermic cover A . The internal energy of the system increases.

From first law,

$$Q_1 = U_2 - U_1 + W_{1-2} \quad (6.14)$$

(for an ideal gas only, $U_1 = U_2$)

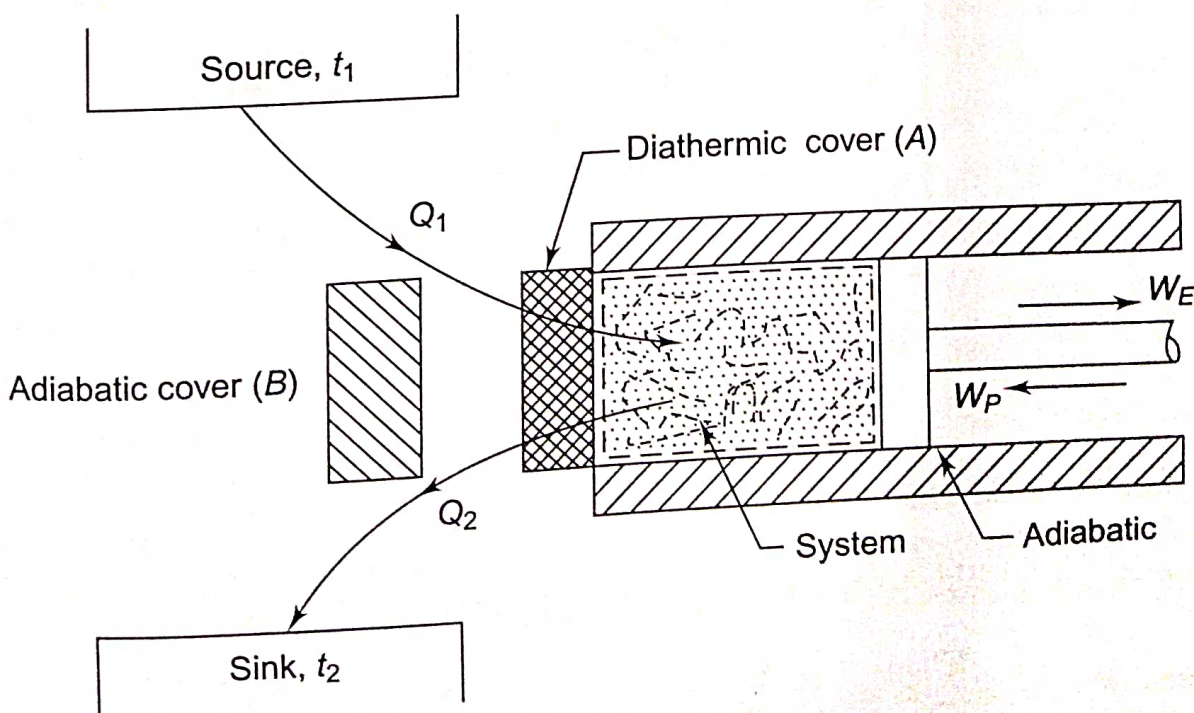


Fig. 6.21

Carnot heat engine—Stationary system

2. *A reversible adiabatic process* in which the diathermic cover A is replaced by the adiabatic cover B , and work W_E is done by the system adiabatically and reversibly at the expense of its internal energy, and the temperature of the system decreases from t_1 to t_2 .

Using the first law,

$$0 = U_3 - U_2 + W_{2-3} \quad (6.15)$$

3. *A reversible isothermal process* in which B is replaced by A and heat Q_2 leaves the system at t_2 to a constant temperature sink at t_2 reversibly, and the internal energy of the system further decreases.

From the first law,

$$-Q_2 = U_4 - U_3 - W_{3-4} \quad (6.16)$$

only for an ideal gas,

$$U_3 = U_4$$

4. *A reversible adiabatic process* in which B again replaces A , and work W_p is done upon the system reversibly and adiabatically, and the internal energy of the system increases and the temperature rises from t_2 to t_1 .

Applying the first law,

$$0 = U_1 - U_4 - W_{4-1} \quad (6.17)$$

Two reversible isotherms and two reversible adiabatics constitute a Carnot cycle, which is represented in p - v coordinates in Fig. 6.22.

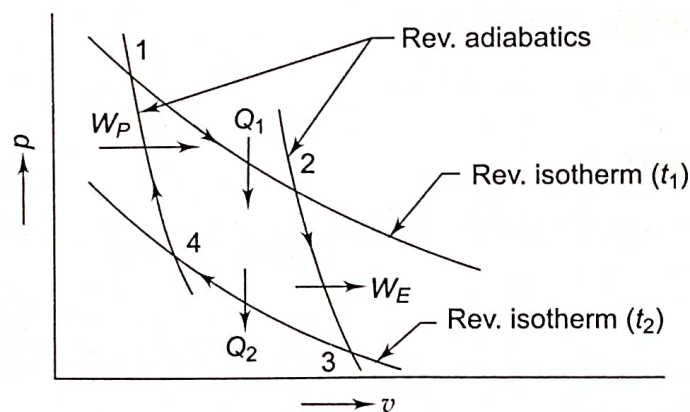


Fig. 6.22 Carnot cycle

Summing up Eqs. (6.14) to (6.17),

$$Q_1 - Q_2 = (W_{1-2} + W_{2-3}) - (W_{3-4} + W_{4-1})$$

or

$$\sum_{\text{cycle}} Q_{\text{net}} = \sum_{\text{cycle}} W_{\text{net}}$$

A cyclic heat engine operating on the Carnot cycle is called a Carnot heat engine.

For a steady flow system, the Carnot cycle is represented as shown in Fig. 6.23. Here heat Q_1 is transferred to the system reversibly and isothermally at t_1 in the

heat exchanger A , work W_T is done by the system reversibly and adiabatically in the turbine (B), then heat Q_2 is transferred from the system reversibly and isothermally at t_2 in the heat exchanger (C), and then work W_P is done upon the system reversibly and adiabatically by the pump (D). To satisfy the conditions for the Carnot cycle, there must not be any friction or heat transfer in the pipelines through which the working fluid flows.

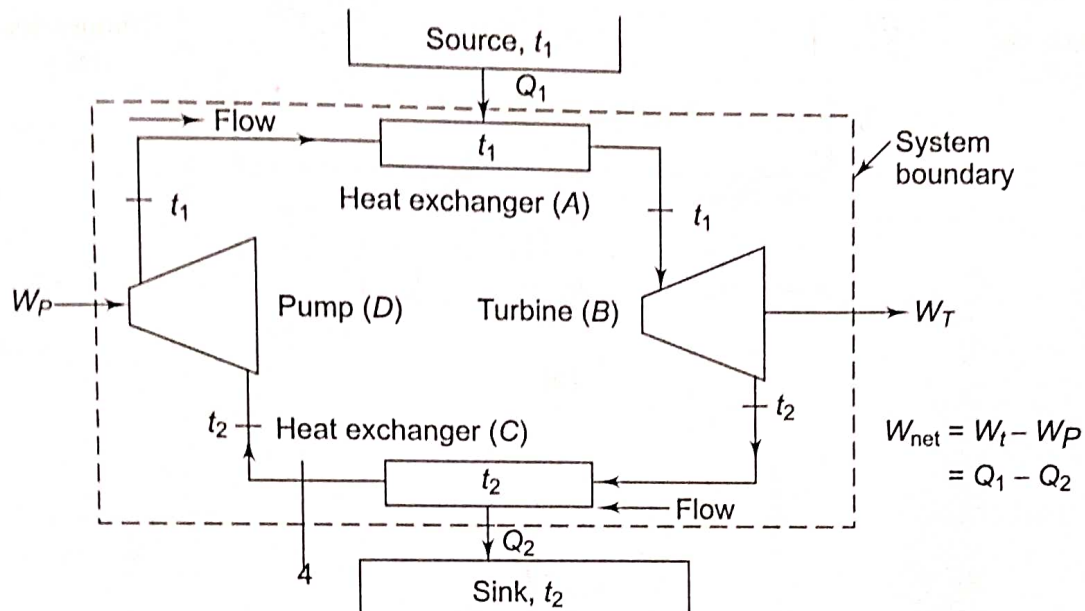


Fig. 6.23 Carnot heat engine—Steady flow system

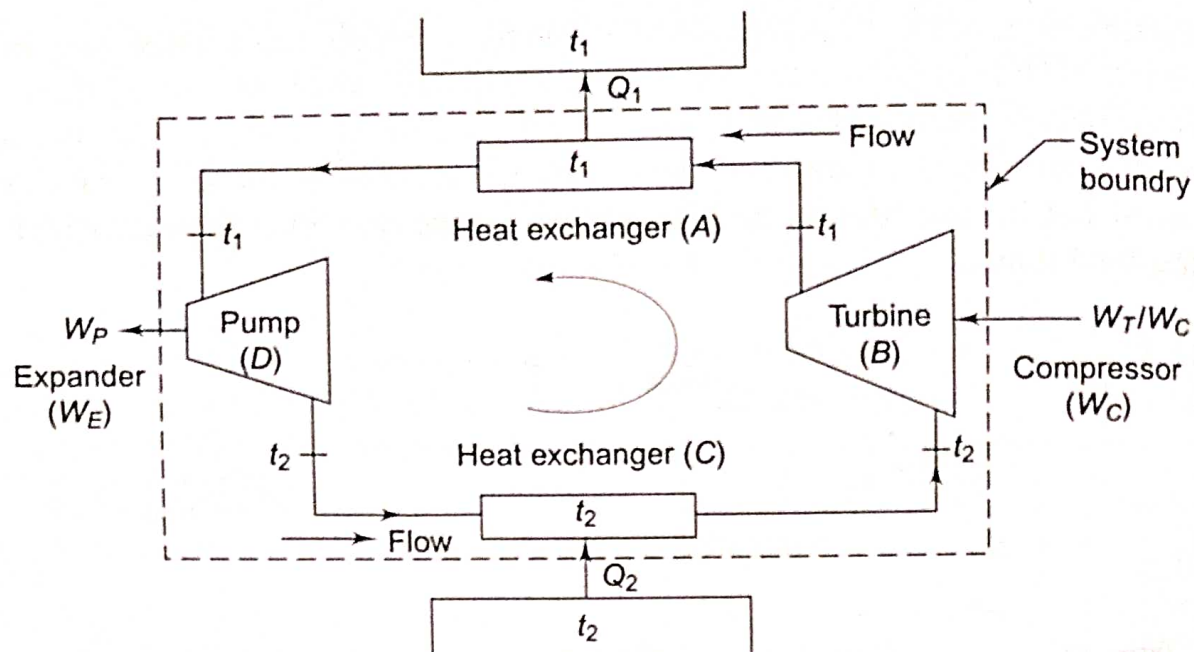
6.13 REVERSED HEAT ENGINE

Since all the processes of the Carnot cycle are reversible, it is possible to imagine that the processes are individually reversed and carried out in reverse order. When a reversible process is reversed, all the energy transfers associated with the process are reversed in direction, but remain the same in magnitude. The reversed Carnot cycle for a steady flow system is shown in Fig. 6.24. The reversible heat engine and the reversed Carnot heat engine are represented in block diagrams in Fig. 6.25. If E is a reversible heat engine (Fig. 6.25a), and if it is reversed (Fig. 6.25b), the quantities Q_1 , Q_2 and W remain the same in magnitude, and only their directions are reversed. The reversed heat engine $\exists$ takes heat from a low temperature body, discharges heat to a high temperature body, and receives an inward flow of network.

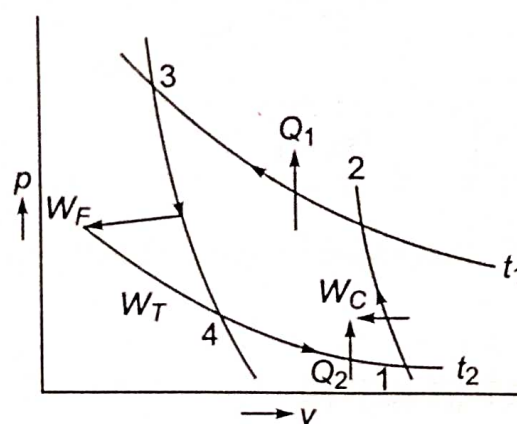
The names *heat pump* and *refrigerator* are applied to the reversed heat engine, which have already been discussed in Section 6.5, where the working fluid flows through the compressor (B), condenser (A), expander (D) and evaporator (C) to complete the cycle.

LO 6.10

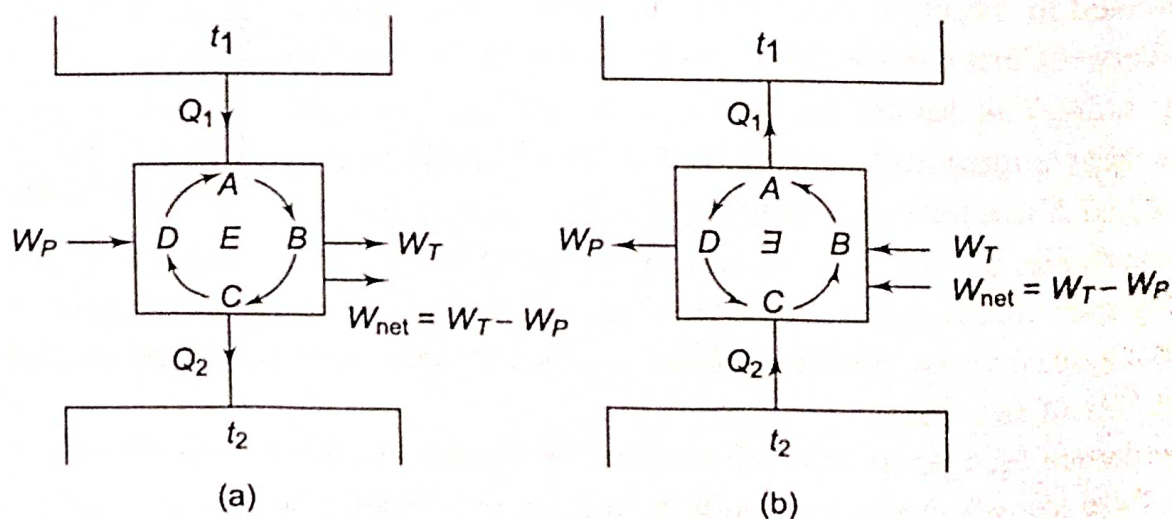
Explain working of Carnot heat pump and refrigerator, and how Carnot theorem is applicable to validate its statement



(a)



(b)

Fig. 6.24 Reversed Carnot heat engine—Steady flow process**Fig. 6.25**

Carnot heat engine and reversed Carnot heat engine shown in block diagrams

6.14 CARNOT'S THEOREM

It states that of all heat engines operating between a given constant temperature source and a given constant temperature sink, none has a higher efficiency than a reversible engine.

Let two heat engines E_A and E_B operate between the given source at temperature t_1 and the given sink at temperature t_2 as shown in Fig. 6.26.

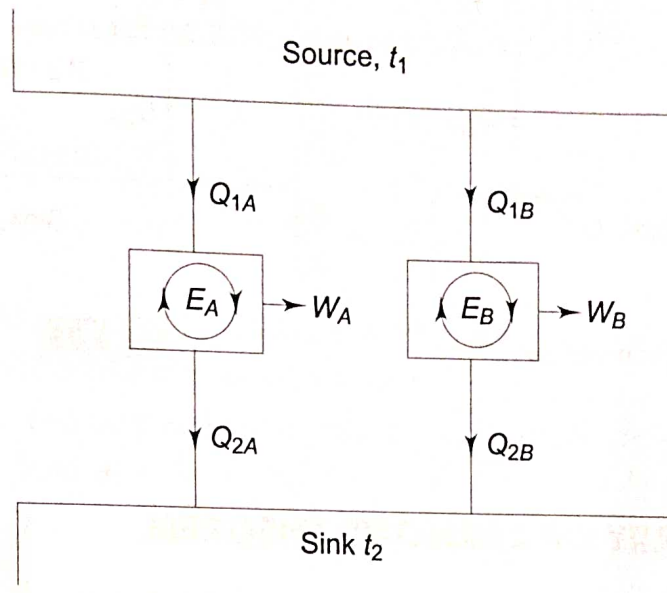


Fig. 6.26 Two cyclic heat engines E_A and E_B operating between the same source and sink, of which E_B is reversible

Let E_A be any heat engine and E_B be any reversible heat engine. We have to prove that the efficiency of E_B is more than that of E_A . Let us assume that this is not true and $\eta_A > \eta_B$. Let the rates of working of the engines be such that

$$Q_{1A} = Q_{1B} = Q_1$$

Since

$$\eta_A > \eta_B$$

$$\frac{W_A}{Q_{1A}} > \frac{W_B}{Q_{1B}}$$

$\therefore$

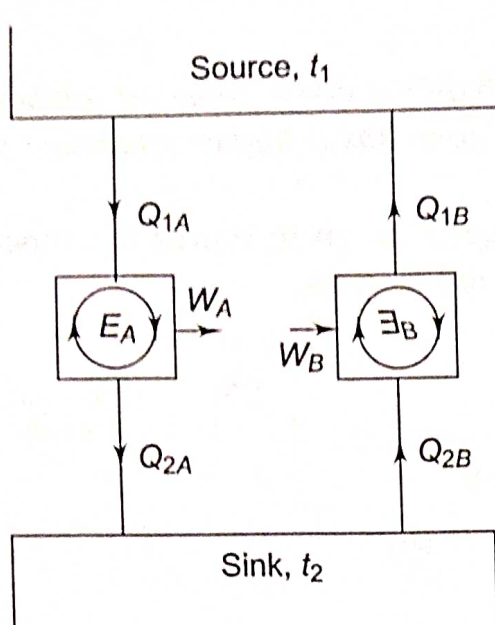
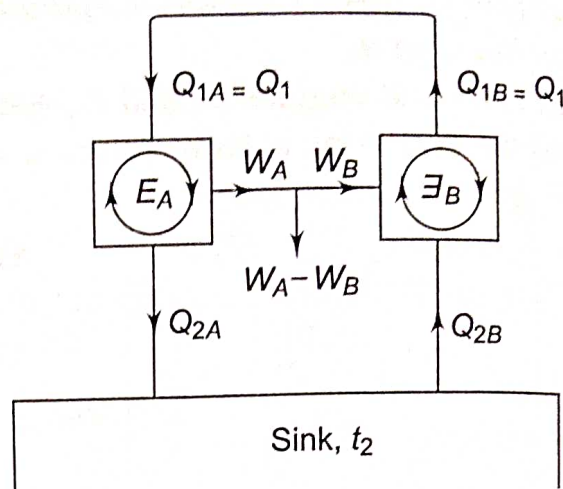
$$W_A > W_B$$

Now, let E_B be reversed. Since E_B is a reversible heat engine, the magnitudes of heat and work transfer quantities will remain the same, but their directions will be reversed, as shown in Fig. 6.27. Since $W_A > W_B$, some part of W_A (equal to W_B) may be fed to drive the reversed heat engine Ξ_B .

Since $Q_{1A} = Q_{1B} = Q_1$, the heat discharged by Ξ_B may be supplied to E_A . The source may, therefore, be eliminated (Fig. 6.28). The net result is that E_A and Ξ_B together constitute a heat engine which, operating in a cycle, produces net work $W_A - W_B$, while exchanging heat with a single reservoir at t_2 . This violates the Kelvin-Planck statement of the second law. Hence the assumption that $\eta_A > \eta_B$ is wrong.

Therefore,

$$\eta_B \geq \eta_A$$

Fig. 6.27 E_B is reversedFig. 6.28 E_A and E_B together violate the K-P statement

6.15 COROLLARY OF CARNOT'S THEOREM

The efficiency of all reversible heat engines operating between the same temperature levels is the same.

Let both the heat engines E_A and E_B (Fig. 6.26) be reversible. Let us assume $\eta_A > \eta_B$. Similar to the procedure outlined in the preceding article, if E_B is reversed to run, say, as a heat pump using some part of the work output (W_A) of engine E_A , we see that the combined system of heat pump, E_B and engine E_A , becomes a PMM2. So η_A cannot be greater than η_B . Similarly, if we assume $\eta_B > \eta_A$ and reverse the engine E_A , we observe that η_B cannot be greater than η_A .

Therefore, $\eta_A = \eta_B$

Since the efficiencies of all reversible heat engines operating between the same heat reservoirs are the same, *the efficiency of a reversible engine is independent of the nature or amount of the working substance undergoing the cycle.*

6.16 THE THERMODYNAMIC TEMPERATURE SCALE

The efficiency of any heat engine cycle receiving heat Q_1 and rejecting heat Q_2 is given by

$$\eta = \frac{W_{\text{net}}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} \quad (6.18)$$

LO 6.11

Develop a point of view from Carnot's theorem and reach at the Third-Law of thermodynamics utilizing absolute thermodynamic temperature scale

By the second law, it is necessary to have a temperature difference ($t_1 - t_2$) to obtain work for any cycle. We know that the efficiency of all heat engines operating between the same temperature levels is the same, and it is independent of the working substance. Therefore, for a reversible cycle (Carnot cycle), the efficiency will depend solely upon the temperatures t_1 and t_2 , at which heat is transferred, or

$$\eta_{\text{rev}} = f(t_1, t_2) \quad (6.19)$$

where f signifies some function of the temperatures. From Eqs. (6.18) and (6.19)

$$1 - \frac{Q_2}{Q_1} = f(t_1, t_2)$$

In terms of a new function F

$$\frac{Q_1}{Q_2} = F(t_1, t_2) \quad (6.20)$$

If some functional relationship is assigned between t_1 , t_2 and Q_1/Q_2 , the equation becomes the definition of a temperature scale.

Let us consider two reversible heat engines, E_1 receiving heat from the source at t_1 , and rejecting heat at t_2 to E_2 which, in turn, rejects heat to the sink at t_3 (Fig. 6.29a).

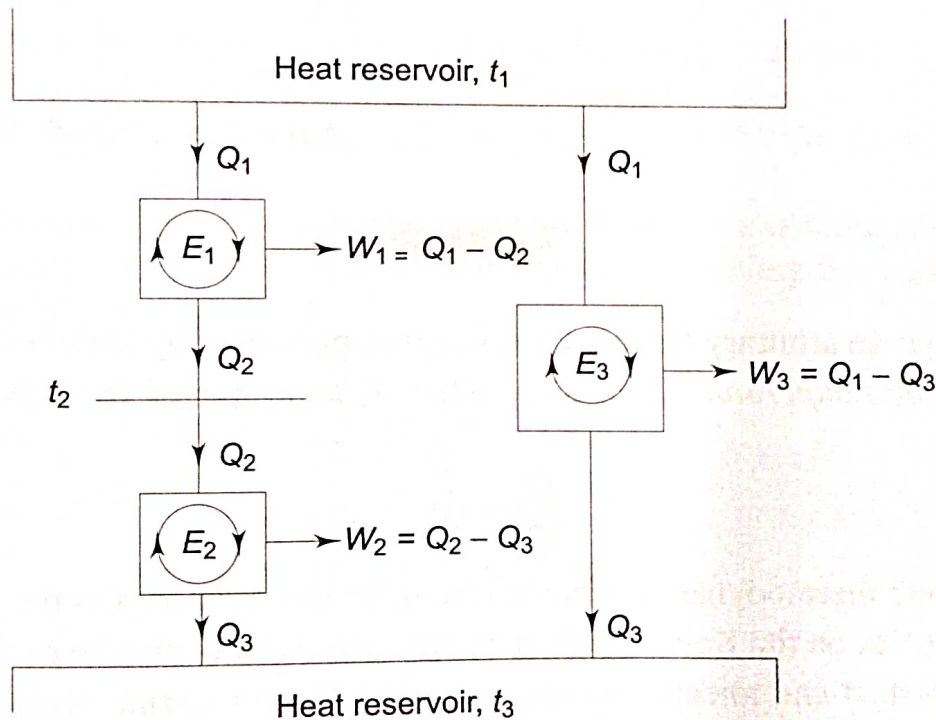


Fig. 6.29(a) Three Carnot engines

Now

$$\frac{Q_1}{Q_2} = F(t_1, t_2); \quad \frac{Q_2}{Q_3} = F(t_2, t_3)$$

E_1 and E_2 together constitute another heat engine E_3 operating between t_1 and t_3 .

$$\therefore \frac{Q_1}{Q_3} = F(t_1, t_3)$$

$$\text{Now} \quad \frac{Q_1}{Q_2} = \frac{Q_1/Q_3}{Q_2/Q_3}$$

$$\text{or} \quad \frac{Q_1}{Q_2} = F(t_1, t_2) = \frac{F(t_1, t_3)}{F(t_2, t_3)} \quad (6.21)$$

The temperatures t_1 , t_2 and t_3 are arbitrarily chosen. The ratio Q_1/Q_2 depends only on t_1 and t_2 , and is independent of t_3 . So t_3 will drop out from the ratio on the right in Eq. (6.21). After it has been cancelled, the numerator can be written as $\phi(t_1)$, and the denominator as $\phi(t_2)$, where ϕ is another unknown function. Thus,

$$\frac{Q_1}{Q_2} = F(t_1, t_2) = \frac{\phi(t_1)}{\phi(t_2)}$$

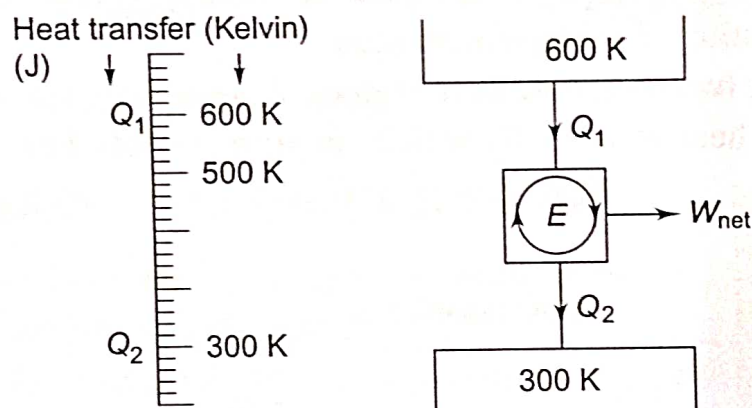


Fig. 6.29(b)

Since $\phi(t)$ is an arbitrary function, the simplest possible way to define the *absolute thermodynamic temperature* T is to let $\phi(t) = T$, as proposed by Kelvin. Then, by definition

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \quad (6.22)$$

The absolute thermodynamic temperature scale is also known as the *Kelvin scale*. Two temperatures on the Kelvin scale bear the same relationship to each other as do the heats absorbed and rejected respectively by a Carnot engine operating between two reservoirs at these temperatures Fig. (6.29 (b)). The Kelvin temperature scale is, therefore, independent of the peculiar characteristics of any particular substance.

The heat absorbed Q_1 and the heat rejected Q_2 during the two reversible isothermal processes bounded by two reversible adiabatics in a Carnot engine can be measured. In defining the Kelvin temperature scale also, the triple point of water is taken as the standard reference point. For a Carnot engine operating between reservoirs at temperatures T and T_t , T_t being the triple point of water (Fig. 6.30), arbitrarily assigned the value 273.16 K,

$$\frac{Q}{Q_t} = \frac{T}{T_t}$$

$\therefore$

$$T = 273.16 \frac{Q}{Q_t} \quad (6.23)$$

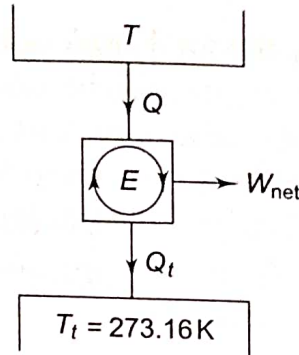


Fig. 6.30 Carnot heat engine with sink at triple point of water

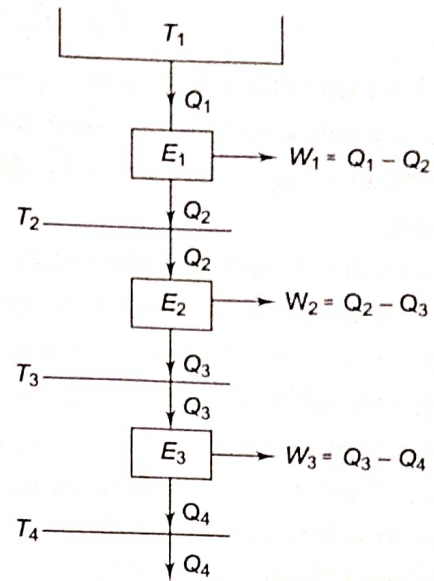


Fig. 6.31 Heat engines operating in series

If this equation is compared with the equations given in Section 2.3, it is seen that *in the Kelvin scale, Q plays the role of thermometric property*. The amount of heat supply Q changes with change in temperature, just like the thermal emf in a thermocouple.

That the absolute thermodynamic temperature scale has a definite zero point can be shown by imagining a series of reversible engines, extending from a source at T_1 to lower temperatures (Fig. 6.31).

Since

$$\frac{T_1}{T_2} = \frac{Q_1}{Q_2}$$

 $\therefore$

$$\frac{T_1 - T_2}{T_2} = \frac{Q_1 - Q_2}{Q_2}$$

or

$$T_1 - T_2 = (Q_1 - Q_2) \frac{T_2}{Q_2}$$

Similarly,

$$T_2 - T_3 = (Q_2 - Q_3) \frac{T_3}{Q_3}$$

$$= (Q_2 - Q_3) \frac{T_2}{Q_2}$$

$$T_3 - T_4 = (Q_3 - Q_4) \frac{T_2}{Q_2}$$

and so on.

If $T_1 - T_2 = T_2 - T_3 = T_3 - T_4 = \dots$, assuming equal temperature intervals

$$Q_1 - Q_2 = Q_2 - Q_3 = Q_3 - Q_4 = \dots$$

or $W_1 = W_2 = W_3 = \dots$

Conversely, by making the work quantities performed by the engines in series equal ($W_1 = W_2 = W_3 = \dots$), we will get

$$T_1 - T_2 = T_2 - T_3 = T_3 - T_4 = \dots$$

at equal temperature intervals. A scale having one hundred equal intervals between the steam point and the ice point could be realised by a series of one hundred Carnot engines operating as in Fig. 6.31. Such a scale would be independent of the working substance.

If enough engines are placed in series to make the total work output equal to Q_1 , then by the first law the heat rejected from the last engine will be zero. By the second law, however, the operation of a cyclic heat engine with zero heat rejection cannot be achieved, although it may be approached as a limit. When the heat rejected approaches zero, the temperature of heat rejection also approaches zero as a limit. Thus it appears that a definite zero point exists on the absolute temperature scale but this point cannot be reached without a violation of the second law.

It follows from Eq. (6.23),

$$T = 273.16 \frac{Q}{Q_i}$$

that the heat transferred isothermally between two given adiabatics decreases as the temperature decreases. Conversely, the smaller the value of Q , the lower the temperature T . The smallest possible value of Q is zero, and the corresponding T is zero. *Thus if a system undergoes a reversible isothermal process without transfer of heat, the temperature at which this process takes place is called absolute zero.* In other words, at absolute zero, an isotherm and an adiabatic are identical.

Also, since $Q_2 = 0$, the isothermal process at $T_2 = 0$ would also be adiabatic rendering Carnot cycle ambiguous.

Thus any attainable value of absolute temperature is always greater than zero. This is also known as the *third law of thermodynamics* which may be stated as follows: *It is impossible by any procedure, no matter how idealised, to reduce any system to the absolute zero of temperature in a finite number of operations.*

This is what is called the Fowler–Guggenheim statement of the third law. The third law itself is an independent law of nature, and not an extension of the second law. The concept of heat engine is not necessary to prove the non-attainability of absolute zero of temperature by any system in a finite number of operations.

6.17 EFFICIENCY OF THE REVERSIBLE HEAT ENGINE

The efficiency of a reversible heat engine in which heat is received solely at T_1 is found to be

$$\eta_{\text{rev}} = \eta_{\text{max}} = 1 - \left(\frac{Q_2}{Q_1} \right)_{\text{rev}} = 1 - \frac{T_2}{T_1}$$

or

$$\eta_{\text{rev}} = \frac{T_1 - T_2}{T_1} \quad (6.24)$$

LO 6.12
Calculate the efficiency of reversible devices and provide the relation between ideal gas temperature and Kelvin temperature

It is observed here that as T_2 decreases, and T_1 increases, the efficiency of the reversible cycle increases.

Since η is always less than unity, T_2 is always greater than zero and positive. The COP of a refrigerator is given by

$$(\text{COP})_{\text{refr}} = \frac{Q_2}{Q_1 - Q_2} = \frac{1}{\frac{Q_1}{Q_2} - 1}$$

For a reversible refrigerator, using

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$[\text{COP}_{\text{refr}}]_{\text{rev}} = \frac{T_2}{T_1 - T_2} \quad (6.25)$$

Similarly, for a reversible heat pump

$$[\text{COP}_{\text{H.P.}}]_{\text{rev}} = \frac{T_1}{T_1 - T_2} \quad (6.26)$$

6.18 EQUALITY OF IDEAL GAS TEMPERATURE AND KELVIN TEMPERATURE

Let us consider a Carnot cycle executed by an ideal gas, as shown in Fig. 6.32.

The two isothermal processes $a-b$, and $c-d$ at θ_1 and θ_2 are represented by equilateral hyperbolas whose equations are, respectively

$$pV = nR\theta_1$$

and

$$pV = nR\theta_2$$

For any infinitesimal reversible process of an ideal gas, the first law may be written as

$$\delta Q = C_v d\theta + pdV = pdV$$

Applying this equation to the isothermal process $a-b$, the heat absorbed is found to be

$$Q_1 = \int_{V_a}^{V_b} pdV = \int_{V_a}^{V_b} \frac{nR\theta_1}{V} dV = nR\theta_1 \ln \frac{V_b}{V_a}$$

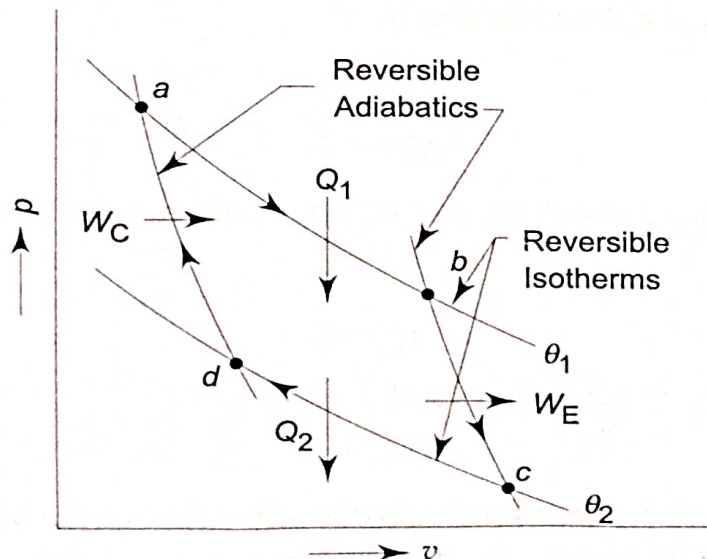


Fig. 6.32 Carnot cycle of an ideal gas

Similarly, for the isothermal process $c-d$, the heat rejected is

$$Q_2 = nR\theta_2 \ln \frac{V_c}{V_d}$$

$$\therefore \frac{Q_1}{Q_2} = \frac{\theta_1 \ln \frac{V_b}{V_a}}{\theta_2 \ln \frac{V_c}{V_d}} \quad (6.27)$$

Since the process $b-c$ is adiabatic, the first law gives

$$-C_v d\theta = pdV = \frac{nR\theta}{V} dV$$

$$\frac{1}{nR} \int_{\theta_2}^{\theta_1} C_v \frac{d\theta}{\theta} = \ln \frac{V_c}{V_b}$$

Similarly, for the adiabatic process $d-a$

$$\frac{1}{nR} \int_{\theta_2}^{\theta_1} C_v \frac{d\theta}{\theta} = \ln \frac{V_d}{V_a}$$

$$\therefore \ln \frac{V_c}{V_b} = \ln \frac{V_d}{V_a}$$

or $\frac{V_c}{V_b} = \frac{V_d}{V_a}$

or $\frac{V_b}{V_a} = \frac{V_c}{V_d} \quad (6.28)$

Equation (6.27) thus reduces to

$$\frac{Q_1}{Q_2} = \frac{\theta_1}{\theta_2} \quad (6.29)$$

Kelvin temperature was defined by Eq. (6.22)

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

If θ and T refer to any temperature, and θ_t and T_t refer to the triple point of water,

$$\frac{\theta}{\theta_t} = \frac{T}{T_t}$$

Since $\theta_T = T_t = 273.16$ K, it follows that

$$\theta = T \quad (6.30)$$

The Kelvin temperature is, therefore, numerically equal to the ideal gas temperature and may be measured by means of a gas thermometer.